SURJECTION $O_X \BOXTIMES F(O_X) \TO K$ WHEN X IS SEPARATED AND FLAT OVER R

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file functors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for functors.tex (fu-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original lemma 0FZH: sources/stacks-chow/tags.txt, source lines 19952–19952. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0GMU: sources/stacks-chow/tags.txt, source lines 20769–20769. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0FZI: sources/stacks-chow/tags.txt, source lines 19953–19953. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02KH: sources/stacks-chow/tags.txt, source lines 3372–3372. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0B55: sources/stacks-chow/tags.txt, source lines 14023–14023. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In Lemma 0FZH let F and $\mathcal{K}$ correspond. If X is separated and flat over R, then there is a surjection $\mathcal{O}_X \boxtimes F(\mathcal{O}_X) \to \mathcal{K}$.*

Proof. Let $\Delta : X \to X \times_R X$ be the diagonal morphism and set $\mathcal{O}_\Delta = \Delta_*\mathcal{O}_X$. Since Δ is a closed immersion have a short exact sequence

$$0 \to \mathcal{I} \to \mathcal{O}_{X \times_R X} \to \mathcal{O}_\Delta \to 0$$

Since $\mathcal{K}$ is flat over X, the pullback $\mathrm{pr}_{23}^*\mathcal{K}$ to $X \times_R X \times_R Y$ is flat over $X \times_R X$. We obtain a short exact sequence

$$0 \to \mathrm{pr}_{12}^*\mathcal{I} \otimes \mathrm{pr}_{23}^*\mathcal{K} \to \mathrm{pr}_{23}^*\mathcal{K} \to \mathrm{pr}_{12}^*\mathcal{O}_\Delta \otimes \mathrm{pr}_{23}^*\mathcal{K} \to 0$$

on $X \times_R X \times_R Y$, see Modules, Lemma 0GMU. Thus, by Lemma 0FZI we obtain a surjection

$$\mathrm{pr}_{13,*}(\mathrm{pr}_{23}^*\mathcal{K}) \to \mathrm{pr}_{13,*}(\mathrm{pr}_{12}^*\mathcal{O}_\Delta \otimes \mathrm{pr}_{23}^*\mathcal{K})$$

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By flat base change (Cohomology of Schemes, Lemma 02KH) the source of this arrow is equal to $\mathrm{pr}_2^*\mathrm{pr}_{2,*}\mathcal{K} = \mathcal{O}_X \boxtimes F(\mathcal{O}_X)$. On the other hand the target is equal to

$$\mathrm{pr}_{13,*}(\mathrm{pr}_{12}^*\mathcal{O}_\Delta \otimes \mathrm{pr}_{23}^*\mathcal{K}) = \mathrm{pr}_{13,*}(\Delta \times \mathrm{id}_Y)_*\mathcal{K} = \mathcal{K}$$

which finishes the proof. The first equality holds for example by Cohomology, Lemma 0B55 and the fact that $\mathrm{pr}_{12}^*\mathcal{O}_\Delta = (\Delta \times \mathrm{id}_Y)_*\mathcal{O}_{X \times_R Y}$. $\square$